

Task-Aware Reuse of Cellular Molecular Receivers: Response Prediction and Feedback-Driven Waiting

Ruifeng Zheng

Deutsche Telekom Chair of Communication Networks

TU Dresden, Germany

Abstract—A living cell used as a molecular receiver remembers earlier commands, so a command sent too early can fail. We ask what information predicts a cell’s reporter response under repeated commands and whether feedback can replace a fixed guard interval. On 1,565 experimental FGF2–ERK single-cell trajectories, reporter filtering plus a calibrated mechanistic (B3) increment beat persistence on the validation and all three test protocols (by 6.7–7.6% in 10-min RMSE); the increment carried this gain (4.5–9.8%; five of six held-out protocols), whereas filtering alone helped on two and input history added at most 2%. In the next-command windows of two test conditions, however, the increment raised the pooled 10-min error by 7.2%, and a later probe in those cells is consistent with B3 overestimating recovery after a long command. In B3 simulations requiring 90% reliability after each previous command (noise s.d. 0.005), waiting on a smoothed reporter or the B3 belief finishes 5.5–7.7 min earlier than history-conditioned fixed waits (38–45% of their gap to an oracle); with a feedback delay the rule must predict readiness, and if B3 is the true receiver the mechanistic estimate pays off at high noise or with 10-min sampling. This speed comes from probing soon after readiness: a 2-min recovery lag hidden from the reporter broke the requirement for both rules but not for fixed waits, and the measured probe implies at least 35.5 min. A delay after the threshold crossing, calibrated on true outcomes, restored it up to 12 min, 6.2–7.7 min faster than recalibrated fixed waits.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, Bayesian filtering, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living cells as receivers offers a natural interface between engineered communication systems and biological processes. Unlike a conventional receiver, a cell retains the effect of earlier chemical stimuli in receptor dynamics, intracellular signaling and feedback, so repeated commands are not independent: a receiver that has just responded may not execute the next command with the same reliability. This raises a communication-level question for repeated molecular tasks: *when can a living receiver be used again, and what information does that decision need?*

Receiver memory is familiar in MC, but it is usually placed in the channel. Diffusion leaves residual molecules that cause intersymbol interference (ISI) [1], [2], and receivers counter it with sequence detection and equalization [3], decoders that exploit memory and sampling rate [4], adaptive thresholds [5], and symbol or detection intervals that trade rate against residual ISI [6], [7]; ligand-binding reception adds its own noise [8]. In these designs the guard interval is set in advance.

For a cellular receiver the memory is a hidden biochemical state seen only through a noisy reporter, and deciding from that reporter when to send the next command is a stopping problem, related to sequential detection [9], [10] and optimal stopping [11]. History-dependent FGF2–ERK responses [12] and frequency-dependent signaling [13] show that such memory exists, feedback control of single cells is feasible [14], [15], and experimental and computational work on biological MC systems is emerging [16]–[19]. What is missing is a measurement-based account of which information predicts a cell’s response under repeated commands, including the next one, and a test of whether feedback can replace a fixed guard interval.

We combine experimental FGF2–ERK single-cell trajectories with the calibrated B3 signaling model of [12]. Nested predictors add the current reporter, causal reporter filters, lagged values, a mechanistic B3 increment and explicit input history one at a time. Receiver reuse is posed as a calibrated stopping rule on B3 with a reliability requirement for each previous command. Two links connect the parts, although the rules do not use the experimental predictors: with a feedback delay the rule must predict readiness ahead, as the forecasts do, and the measured probe that exposes B3’s recovery error after a long command sets the size of the hidden recovery lag under which the waiting rules are stress-tested. Our contributions are:

- On measured cells, reporter filtering plus the B3 increment beats persistence on the validation and all three test protocols (by 6.7–7.6% in 10-min RMSE), and the increment carries this gain: it lowers the 10-min error of the filter on all four (by 4.5–9.8%) and in five of six held-out protocols, and on Test A lifts the noise-adjusted skill at 20–30 min to 21% (9–13% with filtering alone); at 10 min reporter filtering helps on the validation protocol and Test A only, and input history adds at most 1% in RMSE on the test protocols (up to 2% across held-out protocols) and destabilizes longer forecasts on Tests A and B. In the next-command windows of two test conditions the B3 increment raised the pooled 10-min error, through the 25 ng/ml condition, and a later probe pulse in those cells is consistent with B3 overestimating recovery after a long command.
- Under a per-history 90% reliability requirement at the reference noise (s.d. 0.005), waiting on a smoothed reporter or the B3 belief finishes 5.5–7.7 min earlier than history-

conditioned fixed waits and closes 38–45% of their gap to the oracle across three pre-specified calibration targets (42–45% at 0.94, the only target at which every rule met the requirement in all splits); at this noise the smoothed reporter is as good as the B3 belief, and the raw current reporter gains nothing. Because feedback probes soon after readiness, it relies on the model’s recovery timing: under a recovery lag hidden from the reporter, fixed waits and the current-reporter rule kept the requirement up to 2 min and the smoothed and B3-belief rules only without lag, whereas the measured probe implies a lag of at least 35.5 min, at which even the oracle succeeds for only 31% of receivers after the long command; a delay after the threshold crossing, calibrated on true outcomes, restored the requirement up to 12 min at 6.2–7.7 min less completion time than recalibrated fixed waits.

- A condition map over feedback delay, sampling interval and noise shows when a correct mechanistic model is worth using: readiness must be predicted rather than nowcast once the delay reaches about 20 min, and the B3-predictive rule beats a smoothed reporter with its trend by 0.9–5.4 min at high reporter noise (at similar success, at both calibration targets, in 11 of 15 map cells) and by up to 1.1 min with sparse sampling at the reference noise, but not at low noise or with dense sampling at the reference noise; these gains, like those of Contribution 2, presuppose that the model’s recovery timing is right.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ be the commanded FGF2 concentration (ng/ml). Delivery to the receiver is represented abstractly as

$$\begin{aligned} \tilde{u}(t) &= \mathcal{T}_\vartheta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, \tilde{u}), \\ z_j &= g(x_j), \quad y_{j,i} = z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where $\tilde{u}$ is the local concentration, x_j the hidden state of cell j , z_j its noise-free normalized ERK reporter and $y_{j,i}$ the observation at frame t_i with noise $\nu_{j,i}$. Neither $\mathcal{T}_\vartheta$ nor $\tilde{u}$ is calibrated here. The information $\mathcal{H}_t$ holds the executed commands and the observations that have arrived by t .

Reuse task. The amplitude and duration of the next command are fixed; only its timing is decided. A previous command of $m \in \{3, 30\}$ min ends at time 0, and a 5-min probe at the same amplitude acts on the receiver at $\tau \in \{0, 2, \dots, 120\}$ min. With z_j^τ the response when the probe acts at τ and r_j^{nv} the same receiver’s rise to the probe without a previous command,

$$\begin{aligned} S_j(\tau) &= \mathbf{1} \left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_j^{\text{nv}} \right\}, \\ R_{j,t} &= \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t), \end{aligned} \quad (2)$$

with window $W = 20$ min. Success is scored on the noise-free response; decisions see only noisy observations. The completion time is $T = \tau + W$ on success and $T_f = 140$ min

TABLE I
NESTED PREDICTORS AND THEIR FEATURES.

Predictor	u_t, y_t, slope	$\bar{y}_\ell$	lags	$\hat{\delta}_{t,h}$	$v_\ell, \int u$
Persist ($\hat{y}_{t+h} = y_t$)					
Current	✓				
Filter	✓	✓			
Lag	✓		✓		
Filter+B3	✓	✓		✓	
Lag+B3	✓		✓	✓	
F+B3+Input	✓	✓		✓	✓

on failure, the latest completion of any success. The design problem is

$$\min \mathbb{E}[T \mid m] \quad \text{s.t.} \quad \mathbb{P}(S = 1 \mid m) \geq q, \quad m \in \{3, 30\}, \quad (3)$$

with $q = 0.9$ the required reliability, checked empirically on held-out samples rather than guaranteed; rules may be calibrated to higher targets to meet it (Sec. V-C). Requiring q for each previous command, not on average, prevents a rule from buying speed after one history with failures after the other. We use $\kappa = 0.5$, fixed before the waiting analyses after $\kappa = 0.8$ proved infeasible; a later scan over $\{0.30, 0.35, \dots, 0.70\}$ shows that it is also the largest value at which the oracle still succeeds for $> 99\%$ of simulated receivers within 120 min, pooled over both histories (99.25%, 98.5% after the 30-min command; 95.1% at 0.55, 82.0% at 0.60). If the receiver’s state reaches the controller late, or the command takes time to arrive, a probe decided at t acts at $t+d$: the rule must predict R at $t+d$ from $\mathcal{H}_t$, a forecast with horizon $h = d$.

III. PREDICTORS AND WAITING RULES

A. Nested Predictors

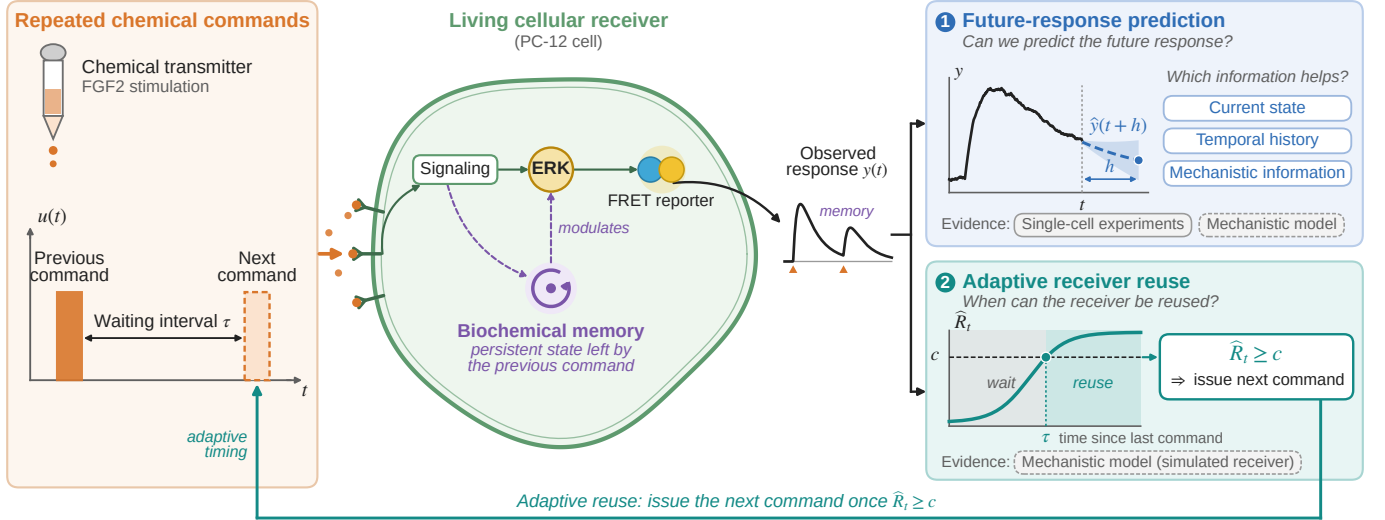
Predictors for horizons $h \in \{2, 10, 20, 30\}$ min start after a fixed 10-min warmup; future reporters are targets only. Table I lists the feature sets (slope: last reporter increment per minute; $\int u$: cumulative command; repository code keys P, C, O, A, M, A+M and H, in table order); every learned predictor also receives the known future command and is fitted by ridge regression with equal weight per condition. Lags are the reporter and command values of the previous 5–30 frames. The causal reporter and input filters with scales $\ell \in \{2, 10, 60\}$ min are

$$\begin{aligned} \bar{y}_{\ell,i} &= e^{-\Delta_i/\ell} \bar{y}_{\ell,i-1} + (1 - e^{-\Delta_i/\ell}) y_i, \\ v_\ell(t) &= \frac{1}{\ell} \int_0^t e^{-(t-s)/\ell} u(s) ds, \end{aligned} \quad (4)$$

with $\bar{y}_{\ell,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$. As mechanism we execute the authors’ B3 model with their 1,000 posterior samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by the cell’s past reporter,

$$\begin{aligned} w_s(t) &\propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \\ \hat{\delta}_{t,h} &= \sum_s w_s(t) [f_s(t+h) - f_s(t)], \end{aligned} \quad (5)$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. Since $\hat{\delta}_{t,h}$



Potential applications: biosensing · closed-loop biochemical control · adaptive therapeutic stimulation · synthetic biological communication

Fig. 1. System concept. Repeated FGF2 commands drive a living receiver whose biochemical memory shapes the ERK/FRET response $y(t)$ to later commands. The response supports (1) prediction of $\hat{y}(t+h)$ and (2) readiness-based reuse, where the next command is sent once $\hat{R}_t \geq c$.

already uses the command history, **F+B3+Input** tests only the extra value of explicit input features. Penalty, σ and lag length are chosen on validation data; normalization, lag filling and fitting details are in the repository.

B. Calibrated Waiting Rules

A waiting rule probes at the first decision time at which its readiness estimate reaches a threshold c , or at 120 min,

$$\tau = \min(\{t : \hat{R}_t \geq c\} \cup \{120\}), \quad (6)$$

a stopping time on the observations, as in sequential tests [9]. Three estimators of R_t are compared on B3: *current* uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; *smoothed* applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the *B3 belief* weights reference receivers under the known command history by all observations, as in Eq. (5), with likelihood s.d. equal to the noise s.d. floored at 0.001. The comparator is one fixed wait per previous command, a history-conditioned guard interval. References are one half of the posterior samples (calibration; each receiver left out of its own references); the other half (evaluation) never enters references, thresholds or waits. Every rule, fixed or adaptive, chooses its parameters per previous command (a wait, a threshold, or for smoothing a scale and a threshold) to minimize that history’s calibration completion subject to its calibration success $\geq$ a target, i.e. Eq. (3) on calibration samples. Twenty splits redraw the halves and the observation noise.

With delay d , the *B3-predictive* rule averages the references’ success at $t+d$ under the same weights, $\hat{R}_{t+d|t} = \sum_s w_s(t) S_s(t+d)$; the *B3 nowcast* ignores the delay. The model-free comparator, *smoothing plus slope*, also extrapolates: a kernel over the smoothed level and its last increment,

$(\bar{y}_t, \bar{y}_t - \bar{y}_{t-\Delta})$, with the covariance of smoothed white noise, maps the current trend to success at $t+d$. Observations arrive every $\Delta \in \{2, 6, 10\}$ min and decisions are made when they arrive.

IV. DATA AND EVALUATION

All numerical files come from the paper-linked repository at a pinned commit [20], [21]; commands follow the authors’ model XML, and per-cell normalization uses only frames before stimulation onset. Table II lists the 22 conditions with their roles and cell counts (1,565 cells in total, 2-min frames). Roles were frozen before any newly used response file was downloaded: Test A is a new protocol, Test B’s timing was inferred from the authors’ simulations, and Test C uses new concentrations recorded in the session of the inspected mixed blocks (role “Viewed”). Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and mixed blocks at 2.5 and 250 ng/ml (“fit” and “arch.” in the last column of Table II), B3 is a calibrated reference, not a model blind to those protocols; it is executed from the authors’ files without refitting (posterior-predictive mean within 8.5×10^{-6} of their exports). In the corrected leave-one-protocol-out (LOPO) each of six protocol groups is held out with predictor tuning restricted to the training protocols. Forecast rows are assigned to the phases stimulation, next command (a pulse starts in $(t, t+h]$), early washout (≤ 30 min after a pulse) and late. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}(h) = \left[\frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{1}{N_{k,h}} \sum_{i \in k} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (7)$$

over the conditions $\mathcal{K}$ of a role. Because every error contains the reporter noise, we also report a noise-adjusted skill,

TABLE II
PROTOCOLS, ROLES AND CELLS.

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	–
Single 60 min	Test B	2.5, 25, 250	167	–
Mixed	Test C	0.25, 25	171	–
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

TABLE III
FORECAST ACCURACY ON MEASURED CELLS.

(a) 10-min RMSE by role

Model	Val.	A	B	C	A, 2 min
Persist	0.02353	0.02287	0.02651	0.02876	0.01302
Current	0.02503	0.02442	0.02715	0.02825	0.01342
Filter	0.02403	0.02220	0.02715	0.02827	0.01283
Lag	0.02343	0.02272	0.02565	0.02769	0.01300
Filter+B3	0.02197	0.02120	0.02448	0.02658	0.01263
Lag+B3	0.02161	0.02114	0.02474	0.02652	0.01270
F+B3+Input	0.02195	0.02098	0.02439	0.02656	0.01260
Filter < Current	4/4	4/4	2/3	1/2	4/4
Filter+B3 < Filter	4/4	3/4	2/3	2/2	3/4
Lag+B3 < Lag	4/4	4/4	3/3	1/2	4/4
F+B3+Input < Filter+B3	1/4	2/4	2/3	1/2	2/4

(b) 10-min RMSE, leave-one-protocol-out

Held-out	Filter	F+B3	F+B3+In	Lag	Lag+B3	B3 use
Sustained	0.02102	0.01969	0.01941	0.02025	0.01931	fit
3/20 pulses	0.03680	0.02886	0.02830	0.03287	0.02837	fit
Single 5 min	0.02243	0.02153	0.02139	0.02196	0.02134	arch.
Single 10 min	0.02084	0.02106	0.02104	0.02116	0.02082	none
Single 60 min	0.02701	0.02410	0.02369	0.02486	0.02422	none
Mixed, 0.25/25	0.02709	0.02623	0.02610	0.02706	0.02628	arch. ^o

(c) Noise-adjusted skill and crossing balanced accuracy (%), Test A

Horizon h (min)	2	10	20	30
Filter	4.0	6.3	12.7	9.0
Filter+B3	8.2	15.5	21.1	21.0
95% interval	6–11	11–19	18–24	17–25
F+B3+Input	8.8	17.4	–1.7	13.5
F+B3 (phase)	3.6	14.8	23.4	23.4
Crossing BA, Filter / F+B3	–	56 / 55	52 / 57	53 / 58

$1 - (\text{MSE} - \hat{\sigma}_\nu^2) / (\text{MSE}_{\text{Persist}} - \hat{\sigma}_\nu^2)$, with $\hat{\sigma}_\nu$ a robust second-difference estimate of white reporter noise per condition (0.006–0.009), and 95% intervals from resampling cells within conditions. Cells share a session and no day identifiers exist, so intervals describe cell sampling, not independent experiments. Every analysis was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

V. RESULTS AND DISCUSSION

A. Which Information Predicts the Response

Table III(a) reports the RMSE of Eq. (7) at 10 min for the validation (Val.) and test roles (A–C) of Table II (last column: 2 min on Test A); bold marks the best predictor per column, and the bottom rows count the conditions in which

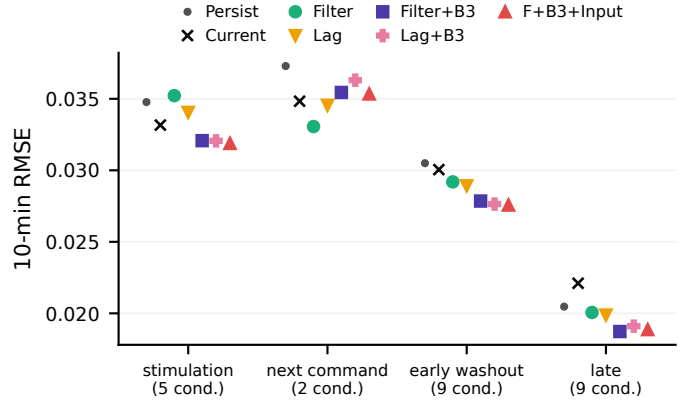


Fig. 2. Condition-equal 10-min RMSE on Tests A–C by forecast phase (axis not starting at zero).

the first predictor of each pair has the lower MSE. The most robust result is that Filter+B3 beat persistence on all four roles, by 6.7–7.6%, and that the B3 increment carried this gain: it lowered the error of Filter on every role (4.5–9.8%), whereas filtering helped only on the validation protocol and Test A (4.0% and 9.1%; none on Tests B and C at 10 min, although at 2, 20 and 30 min it helped on all four), and Filter beat persistence only on Tests A and C (by 2.9% and 1.7%) because Current was itself worse than Persist on three roles. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to Current, the B3 increment by 4.5% relative to Filter, and input history by 1.0% relative to Filter+B3, giving 0.02098, 8.2% below Persist; these reductions use different baselines and are not additive. On the MSE scale the three Test A reductions are 17.3%, 8.9% and 2.0%; subtracting the white-noise floor, under a tenth of the Persist MSE at 10 min, raises them only to 18.8%, 9.8% and 2.2%. Filter+B3 explains 15.5% of the noise-free Persist error (Table III(c) gives this noise-adjusted skill by horizon, with a 95% cell-bootstrap interval for Filter+B3), and the cell-bootstrap intervals of the three RMSE reductions were 7.4–10.9%, 3.0–6.1% and 0.6–1.4%. The input-history gain was small and not consistent across conditions (two of four on Test A; 0.4% and 0.1% on B and C). The lagged baseline Lag was 2.3% worse than Filter on Test A, but adding the B3 increment to it still reduced the error by 6.9% (four of four concentrations), so on Test A the mechanistic gain holds for both the filter and the lag representation of the history. In the corrected LOPO (Table III(b), where F+B3 and F+B3+In abbreviate Filter+B3 and F+B3+Input; bold: best per held-out protocol; last column: B3's prior use of that protocol, with ^o marking use at other concentrations), F+B3+Input beat Filter+B3 in all six folds (by 0.1–2.0%) and Lag+B3 beat Lag in all six, but with the 10-min pulse held out, one of the two protocols unseen by B3, Filter+B3 was 1.1% worse than Filter. At longer horizons the mechanistic increment mattered more: on Test A its skill rose to 21% at 20 and 30 min against 9–13% for Filter, and it anticipated the half-decay crossing of the reporter more often

TABLE IV
WAITING ON B3 WITH PER-HISTORY CALIBRATION TO 0.94.

Rule	S_3	T_3	S_{30}	T_{30}	ΔT	G (%)	Met
Fixed per history (94/110)	95.0	115.3	95.6	130.4	–	–	20/20
Current obs.	95.0	116.5	95.4	128.2	+0.2	–1	20/20
Smoothed	94.0	108.4	94.6	119.6	–7.7	45	20/20
B3 belief	95.4	109.0	95.0	119.9	–7.2	42	20/20
Oracle	100.0	99.0	98.4	110.4	–	100	–
Noise s.d. 0.01 (same fixed waits)							
Smoothed	94.4	111.8	93.6	122.5	–4.4	26	20/20
B3 belief	95.4	110.5	94.4	121.1	–6.0	35	19/20

(balanced accuracy 57–58% versus 52–53%, “Crossing BA” in Table III(c); 65–67% versus 54–57% on Test B), although no predictor did so reliably. Explicit input history became harmful on Tests A and B: $F+B3+Input$ fell to -1.7% skill at 20 min on Test A and to -50% and -53% on Test B at 20 and 30 min, although on Test C it lowered the 20- and 30-min error by 3.6% and 3.0%.

B. The Next-Command Window

By phase (Fig. 2), the B3 increment reduced the error of Filter during stimulation (8.9%), early washout (4.6%) and the late phase (6.6%), but raised it in the next-command window from 0.03306 to 0.03544; at 2, 20 and 30 min it lowered the next-command error (by 2.9%, 4.0% and 6.9%), so the failure is specific to the 10-min horizon. That window holds two mixed conditions with ten forecast origins each, five before the 30-min pulse (20 min after the initial 3-min pulse) and five before the 5-min pulse that follows it after 60 min: the increment helped at 0.25 ng/ml (0.03652 to 0.03485) and hurt at 25 ng/ml (0.02920 to 0.03603). Two phase-gated variants fitted on the training protocols did not repair this: a switch to Filter in next-command rows was rejected by the training data, where the increment helped (in-sample MSE 6.3×10^{-4} versus 8.4×10^{-4}), and phase-specific weights ($F+B3$ (phase) in Table III(c)) raised the next-command error to 0.03571 (held-out mixed fold: 0.03630, versus 0.03537 for Filter+B3 and 0.03302 for Filter). The training protocols contain no probe after a long command, so their next-command windows cannot teach the correction. Splitting the window by the following pulse (a frozen descriptive analysis) shows where the increase arose. At 25 ng/ml the increment raised the error before the 5-min pulse that follows the 30-min pulse by 35% (0.02737 to 0.03694; worse in 82 of 84 cells): it predicted a rise (mean $+0.013$) where the reporter fell (-0.009), so its forecasts were too high (mean error $+0.022$ versus $+0.015$ for Filter). Before the 30-min pulse it raised the error by 13.5% (0.03092 to 0.03510) with a slightly negative mean error, as at the first 3/20 probe. The window after the long command thus carried 69% of the increase at 25 ng/ml, below the pre-specified 75% needed to confine it there; the over-prediction is consistent with B3 overestimating recovery after the long command (Sec. V-E).

C. Waiting under a Per-History Reliability Requirement

Readiness and waiting were simulated on B3 with 25 ng/ml commands and observations every 2 min. Readiness depended almost only on the current reporter, waits of 6–48 min never succeeded, and no state met the criterion without the probe. In Fig. 3(a,b) the smoothed and B3-belief frontiers lie between the fixed waits and the oracle and nearly coincide, whereas the current-observation frontier is no better than the fixed waits after the 3-min command and better only at 90% success or above after the 30-min command. Calibrated to 0.90, every rule, fixed waits included, met 0.9 for both histories in only 7–11 of 20 splits, so a calibration target equal to the requirement leaves no margin. At 0.94 all rules met it in every split at noise 0.005, which is why Table IV tabulates this target, a choice made after the splits were evaluated. The table gives, for each rule (“Current obs.”: the current estimator), the evaluation success S_m (%) and mean completion T_m (min) after the m -min command on the primary split at the reference noise s.d. 0.005 (lower block: 0.01 with the same fixed waits) and, as medians over the 20 splits, the difference $\Delta T = T_{\text{rule}} - T_{\text{fix}}$ in mean completion T over both histories (negative: earlier), the closed share $G = (T_{\text{fix}} - T_{\text{rule}})/(T_{\text{fix}} - T_{\text{oracle}})$ of the gap to the oracle, and the number of splits (Met) with $S_3, S_{30} \geq 90\%$. Feedback finished 7.2 min (B3 belief) and 7.7 min (smoothed) earlier than per-history fixed waits (94 and 110 min on the primary split), faster in all 20 splits, and closed 42% and 45% of the gap to the oracle. At the other pre-specified targets, 0.90 and 0.92, where the requirement was met in 7–11 and 16–19 of 20 splits, feedback closed 38–42% of the gap (saving 5.5–6.6 min), again faster in all 20 splits. The current-reporter rule closed none of it. Under a pooled requirement (success averaged over both histories), the B3 belief and smoothing closed 33% and 38% of the gap between pooled history-conditioned waits and the oracle (4.8 and 5.5 min; primary split). At noise s.d. 0.01 the B3 belief beat smoothing in all splits at every target (split medians 1.5–1.7 min; 30–35% versus 21–26% of the gap), whereas at 0.005 smoothing was 0.2–0.4 min faster. Larger failure penalties ($T_f = 160$ –260 min, target 0.90) raised the B3 belief’s gain over fixed waits from 5.5 to 5.9–8.7 min, with the smoothed and B3-belief rules still ahead of the fixed waits. Sensitivity analyses with pooled calibration (repository) agree: the gain persists at $\kappa = 0.4$ and at a shared calibration margin, at $\kappa = 0.6$ even the oracle fails for about 18% of receivers, and under correlated noise the B3 belief needs a likelihood that models the correlation (median success over 20 splits 81% with frozen thresholds and 92% with an AR(1) likelihood, which met 0.9 in 13 of 20 splits), whereas the smoothed rule recovers by recalibration alone (19 of 20).

D. Delay, Noise and Sampling: When the Mechanism Pays Off

With delay, readiness must be predicted. With every rule calibrated to 0.94 per history, as in Table IV, the B3 nowcast became slower than the fixed waits from $d = 20$ min on (124–138 versus 122 min), whereas both predictive rules lost only about 4 min between $d = 0$ and 40 min (Fig. 3(c)), so the

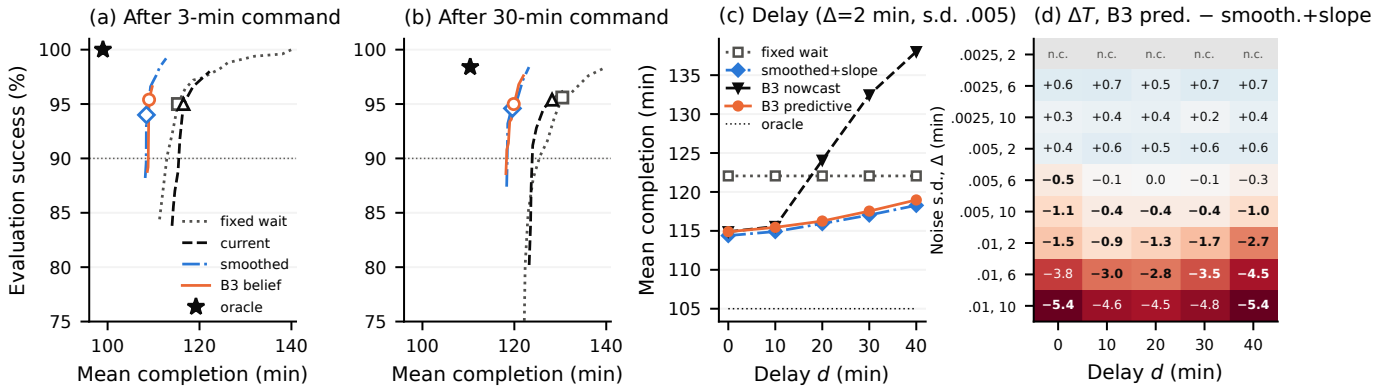


Fig. 3. Waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a,b) Reliability–latency frontiers on evaluation samples (primary split, noise s.d. 0.005; thresholds or waits swept, descriptive), with the operating points calibrated to 0.94 (open markers). (c) Median completion over 20 splits versus feedback delay d (calibration target 0.94 per history). (d) Condition map: median completion of the B3-predictive minus the smoothed+slope rule (min; negative: the model is faster) over delay, noise s.d. and sampling interval Δ ; bold: model faster in ≥ 16 of 20 splits at no more than 0.5 points lower success at targets 0.94 and 0.90; n.c.: the B3 rule never reached the target. One simulated model structure, not measured cells.

value of feedback over fixed waits shrank from 7.2–7.9 to 3.1–3.8 min. The condition map (Fig. 3(d)) answers when the model is needed. At noise s.d. 0.01 the B3-predictive rule was faster than smoothing plus slope in every split of every cell, by 0.9–5.4 min, and the model-free rule became slower than the fixed waits with 10-min sampling and, from $d = 20$ min, with 6-min sampling (by 0.5–4.7 min), while the B3 rule still beat them (by 1.0–3.8 min). At the reference noise the model helped with 10-min sampling at every delay (0.4–1.1 min) and with 6-min sampling only without delay (0.5 min); with 2-min sampling and at noise 0.0025 the smoothed trend was 0.2–0.7 min faster. At noise 0.0025 with 2-min sampling the B3-predictive rule never reached the target on calibration (median success 87.6–90.2%), consistent with weight collapse at low noise, so these five cells are not classified; elsewhere the fixed waits met the requirement in all 20 splits and the B3-predictive and smoothing-plus-slope rules in 19 or 20. The model was faster in at least 16 of 20 splits in 21 of the 45 cells and also no more than 0.5 points below the smoothing-plus-slope rule's median success in 18 of them; the other three, all at noise 0.01, missed this by 0.6–1.6 points. Seventeen of the 18 were also selected at target 0.90 (repository) and are bold in Fig. 3(d); the class of two cells (0.005/6/40 and 0.01/10/30) depends on the target. The success condition, on pooled median success, limits but does not rule out trading reliability for speed after one history.

E. Natural Probes in Measured Cells

Two recorded protocols contain later pulses that act as probes at fixed waits (25 ng/ml): in the 3/20 protocol each 3-min pulse comes 20 min after the previous one, with the cell's own first-pulse rise as naive reference (58 cells); in the mixed protocol (84 cells) a 5-min pulse comes 60 min after a 30-min pulse, scored against single 5-min pulse cells (92 cells). Both probes were early by B3's standard: B3 predicted no success for any sample, and every estimator called no cell ready. The measured cells agreed: 19% met the criterion at the

first 3/20 probe and 3–9% at later ones, and none at the mixed probe, so the rules made no false ready call but missed the 19%. The rises differed, however: at the mixed probe measured cells reached a median 4% of the naive rise where B3 predicts 31%, consistent with B3 overestimating recovery after a long command. These cells also supply the 25 ng/ml next-command window, in which the increment over-predicted the reporter before this probe (Sec. V-B), so the probe and that forecast error are not independent evidence. After a short pulse B3 erred the other way: at the first 3/20 probe it underestimated the rise (8% versus 16% of the naive rise). The B3 weights collapsed on measured cells (effective sample size 1.0–1.4), so a real-data filter needs process noise or cell heterogeneity, and the measured noise floor (0.007–0.008) lies between the map's reference and high noise, where with 2-min sampling the map turns from the smoothed trend to the model, which would help only if it were right (Sec. V-F).

F. When the Model Misjudges Recovery

Every waiting rule above, fixed waits included, is calibrated on B3 outcomes, so none is model-free, and Sec. V-E suggests that B3 recovers too fast after a long command. A frozen simulation therefore delayed the outcomes after the 30-min command by a lag L that the reporter does not show (target 0.94, noise 0.005, 20 splits). Fixed waits and the current-reporter rule kept the per-history requirement up to $L = 2$ min (in 20 and 17 of 20 splits), whereas the smoothed and B3-belief rules failed it at 2 min (10 and 15 of 20), and at $L = 4$ min their success after the 30-min command fell from 95% to 84–85%, against 94% to 89% for the fixed waits: feedback saves time by probing soon after readiness, which leaves little slack for a recovery error that the reporter hides. The measured mixed probe corresponds to a lag of at least 35.5 min (its 4% rise lies within the interquartile range of rises expected from noise alone), at which even the oracle succeeds for only 31% of receivers after the 30-min command; these are the cells of Sec. V-E and of the next-

command window, not independent evidence. Recalibrating only the thresholds on true outcomes restored the feedback rules up to 4–6 min, but calibrating a delay after the threshold crossing together with the threshold restored them in 18–20 of 20 splits up to $L = 12$ min (recalibration was feasible in every split up to 10 min and in 12 at 12 min), 6.2–7.7 min faster than recalibrated fixed waits; this uses the complete outcome curves of the calibration receivers and a common lag, so it is an upper bound. When the reporter shows the lag, the B3 belief was the less robust rule: at noise 0.01 it lost more success than smoothing in 17 and 19 of 20 splits at 12 and 16 min, as its effective sample size fell to 16 and 6.7 (no consistent difference at 4 and 8 min), and at 0.005 in every split from 8 min. Within B3, for the slowest quarter of receivers (readiness onset ≥ 98 min after the 30-min command), only the current rule met the requirement in any split (7 of 20), but the B3-belief and smoothed rules raised success after the 30-min command above that of the fixed waits (by 5.5 and 7.1 points, in 18 and 16 of 20 splits): feedback absorbs variability that the reporter shows, not a recovery error that it hides.

G. Implications for MC Design

A previous command acts on the next one like ISI whose memory sits in the receiver’s biochemistry rather than only in the channel [2], [6]. A fixed wait is a guard interval; conditioning it on the known previous command is the first step (94 and 110 min after the 3- and 30-min commands at target 0.94, primary split), and requiring reliability for each previous command rather than on average lengthens the guard after the long command (106 instead of 102 min at target 0.90, primary split). Feedback then adapts to state variability within each history, much as adaptive detection adapts to residual interference [4], [5], and recovers about 40% of the gap to the oracle at the reference noise. Which estimator feedback needs depends on the noise, the sampling interval and the loop delay: a smoothed reporter with its trend suffices for clean feedback and for frequent feedback at the reference noise, whereas a model is worth its cost for noisy feedback and for sparse feedback at the reference noise, provided it is right about recovery; with a loop delay, either estimate must be predictive. Feedback, however, spends the slack that a guard interval keeps: calibrated on B3, it tolerated no hidden recovery error, so a reusable cellular receiver needs a model that is right about recovery or a guard after the trigger calibrated on probe outcomes (Sec. V-F). The two evidence streams meet at this point: on measured cells the mechanistic increment mattered most at 20–30-min horizons (Table III(c)), and on B3 a loop delay of about 20 min or more requires a predictive rather than a nowcast rule, so a mechanistic forecast is most useful where the control loop is slow, provided it is right about recovery, which after a long command it was not on the measured cells (Secs. V-B, V-E and V-F). A prospective test should randomize 3- versus 30-min preconditioning per chamber, probe at 60–120 min, measure the naive response in same-session controls, calibrate the guard after the trigger on these probe outcomes and count sessions rather than cells.

H. Evidence Boundaries

Data. The prediction comparisons are post hoc on inspected roles; in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%; Test C shares a session with inspected data, no experimental-day identifiers exist, protocol generalization is therefore not cross-day generalization, and B3 had seen some protocols; the noise floor assumes white reporter noise. *Model.* Waiting results come from one simulated model structure whose posterior samples express parameter uncertainty, not validated cell heterogeneity; the 20 splits reuse one finite posterior set; the B3 rules share the plant’s structure, which favors them; the recovery errors are stylized time shifts common to all receivers, and the delay recalibration assumes complete outcome curves; the targets have no coverage guarantee and a calibration margin is not one. *Task.* $\kappa = 0.5$ was adopted after $\kappa = 0.8$ proved infeasible; the naive response is a simulation counterfactual; amplitude and duration were fixed; the natural probes cover one dose at two fixed waits and score readiness calls, not completion time; the experimental predictors were not tested as decision models, so forecast gains do not imply better waiting decisions.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering plus a fixed calibrated mechanistic increment beat persistence on the validation and all three test protocols, with the increment carrying the gain (it also helped in five of six held-out protocols), whereas filtering alone helped on two of the four and explicit input history added little; the increment raised the pooled 10-min error in the next-command windows of two test conditions, through the 25 ng/ml condition, and a later probe pulse in those cells is consistent with B3 overestimating recovery after a long command. In B3 simulations with a reliability requirement for each previous command, waiting on a smoothed reporter or the B3 belief closes about 40% of the gap between history-conditioned guard intervals and the oracle at the reference noise; a smoothed reporter with its trend suffices for clean feedback and for frequent feedback at the reference noise, a correct mechanistic model pays off for noisy feedback and for sparse feedback at the reference noise, and with a feedback delay of about 20 min or more, readiness must be predicted rather than nowcast. Feedback saves this time by probing soon after readiness, so it tolerated almost no recovery error that the reporter hides, whereas fixed waits keep slack; a delay after the trigger, calibrated on outcomes, restored the requirement in simulation while keeping most of the gain, so reusing a cellular receiver needs measured probe outcomes, not only reporter traces, to set that delay.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, implementation details and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [12], [20],

[21]; author files keep their attribution and receive no new license.

REFERENCES

- [1] N. Farsad, H. B. Yilmaz, A. Eckford, C.-B. Chae, and W. Guo, "A comprehensive survey of recent advancements in molecular communication," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 3, pp. 1887–1919, 2016, doi: 10.1109/COMST.2016.2527741.
- [2] V. Jamali, A. Ahmadzadeh, W. Wicke, A. Noel, and R. Schober, "Channel modeling for diffusive molecular communication—a tutorial review," *Proceedings of the IEEE*, vol. 107, no. 7, pp. 1256–1301, 2019, doi: 10.1109/JPROC.2019.2919455.
- [3] D. Kilinc and O. B. Akan, "Receiver design for molecular communication," *IEEE Journal on Selected Areas in Communications*, vol. 31, no. 12, pp. 705–714, 2013, doi: 10.1109/JSAC.2013.SUP2.1213003.
- [4] R. Mosayebi, H. Arjmandi, A. Gohari, M. Nasiri-Kenari, and U. Mitra, "Receivers for diffusion-based molecular communication: Exploiting memory and sampling rate," *IEEE Journal on Selected Areas in Communications*, vol. 32, no. 12, pp. 2368–2380, 2014, doi: 10.1109/JSAC.2014.2367732.
- [5] M. Damrath and P. A. Hoehner, "Low-complexity adaptive threshold detection for molecular communication," *IEEE Transactions on NanoBioscience*, vol. 15, no. 3, pp. 200–208, 2016, doi: 10.1109/TNB.2016.2520566.
- [6] B. Tepekule, A. E. Pusane, H. B. Yilmaz, C.-B. Chae, and T. Tugcu, "ISI mitigation techniques in molecular communication," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 1, no. 2, pp. 202–216, 2015, doi: 10.1109/TMBMC.2015.2501745.
- [7] T. N. Cao, N. Zlatanov, P. L. Yeoh, and J. S. Evans, "Optimal detection interval for absorbing receivers in molecular communication systems with interference," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 6, no. 3, pp. 184–198, 2020, doi: 10.1109/TMBMC.2020.3027250.
- [8] M. Pierobon and I. F. Akyildiz, "Noise analysis in ligand-binding reception for molecular communication in nanonetworks," *IEEE Transactions on Signal Processing*, vol. 59, no. 9, pp. 4168–4182, 2011, doi: 10.1109/TSP.2011.2159497.
- [9] A. Wald, "Sequential tests of statistical hypotheses," *The Annals of Mathematical Statistics*, vol. 16, no. 2, pp. 117–186, 1945, doi: 10.1214/aoms/1177731118.
- [10] T.-Y. Tung and U. Mitra, "Robust molecular communications: DFE-SPRTs and synchronisation," in *Proc. IEEE International Conference on Communications (ICC)*, Shanghai, China, 2019, pp. 1–6, doi: 10.1109/ICC.2019.8761054.
- [11] H. V. Poor and O. Hadjiladis, *Quickest Detection*. Cambridge, UK: Cambridge University Press, 2008, doi: 10.1017/CBO9780511754678.
- [12] Y. Blum, J. Mikelson, M. Dobrzyński, H. Ryu, M.-A. Jacques, N. L. Jeon, M. Khammash, and O. Pertz, "Temporal perturbation of ERK dynamics reveals network architecture of FGF2/MAPK signaling," *Molecular Systems Biology*, vol. 15, no. 11, p. e8947, 2019, doi: 10.15252/msb.20198947.
- [13] H. Ryu, M. Chung, M. Dobrzyński, D. Fey, Y. Blum, S. S. Lee, M. Peter, B. N. Kholodenko, N. L. Jeon, and O. Pertz, "Frequency modulation of ERK activation dynamics rewires cell fate," *Molecular Systems Biology*, vol. 11, p. 838, 2015, doi: 10.15252/msb.20156458.
- [14] L. Postiglione, S. Napolitano, E. Pedone, D. L. Rocca, F. Aulicino, M. Santorelli, B. Tumaini, L. Marucci, and D. di Bernardo, "Regulation of gene expression and signaling pathway activity in mammalian cells by automated microfluidics feedback control," *ACS Synthetic Biology*, vol. 7, no. 11, pp. 2558–2565, 2018, doi: 10.1021/acssynbio.8b00235.
- [15] J.-B. Lugagne, C. M. Blassick, and M. J. Dunlop, "Deep model predictive control of gene expression in thousands of single cells," *Nature Communications*, vol. 15, p. 2148, 2024, doi: 10.1038/s41467-024-46361-1.
- [16] N. Ntetsikas, S. Kyriakoudi, A. Kirmizis, I. Krikidis, I. F. Akyildiz, and M. Lestas, "Noise characterization and robust signal detection in yeast pheromone molecular communication," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 11, no. 2, pp. 218–227, 2025, doi: 10.1109/TMBMC.2025.3554640.
- [17] M. Scherer, L. Brand, L. Wolf, T. tom Dieck, M. Schäfer, S. Lotter, A. Burkovski, H. Sticht, R. Schober, and K. Castiglione, "Closed-loop long-term experimental molecular communication system," *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 12, pp. 22–41, 2026, author version: arXiv:2502.00831v2, doi: 10.1109/TMBMC.2025.3615924.
- [18] M. G. Durmaz, N. Tulluk, R. D. Aksoy, H. B. Yilmaz, B. Yang, A. Wipat, A. E. Pusane, G. Mısırlı, and T. Tugcu, "BioRxToolbox: a computational framework to streamline genetic circuit design in molecular data communications," *Synthetic Biology*, vol. 9, no. 1, p. ysae015, 2024, doi: 10.1093/synbio/ysae015.
- [19] M. T. Barros and S. Dey, "Feed-forward and feedback control in astrocytes for Ca^{2+} -based molecular communications nanonetworks," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 17, no. 4, pp. 1174–1186, 2020, doi: 10.1109/TCBB.2018.2887222.
- [20] LFNS_MSB contributors, "LFNS_MSB: Paper-linked experimental time courses and inference exports," GitHub, branch MSB_version, 2019, pinned commit 5c917ab. https://github.com/Mijan/LFNS_MSB.
- [21] M. Dobrzyński, "Temporal perturbation of Erk dynamics reveals network architecture of FGF2-MAPK signaling," Mendeley Data, version 2, 2020, contributor: Y. Blum. doi: 10.17632/ccnxn84w8z.2.